

UEC735: BIG DATA ANALYTICS

L	T	P	Cr
2	0	2	3.0

Course Objective: Big Data Analytics course will inspire students to explore opportunities in the world of big data analytics. This course will take you from the basics of big data analytics to the advance analytical tools, methods and technology, which could be used for the big data analytics projects. It also brings together several key big data technologies used for storage, analysis and manipulation of data.

Introduction to big data: Introduction to Big Data Platform – Challenges of Conventional Systems - Intelligent data analysis – Nature of Data - Analytic Processes and Tools - Analysis vs. Reporting, Use cases

Big-data Characteristics and issues

Characteristics and issues in Big-Data Analytics, Challenges associated with Big-data, Bog-Data Analytical platforms, Storage and Architecture properties,

Big data using Apache Hadoop Stack

Introduction to Hadoop, HDFS and its architecture, Hadoop-Python framework and programming, Parallelization, MapReduce, Hadoop Client, Apache Sqoop Apache Flume, Hadoop Security, Apache Spark, Spark-Python framework, Pyspark programming and applications

Apaxe Hadoop Tools: Overview of hive and its architecture, Hive data types and File format, Hive query language (HQL), Apaxe Storm, Introduction to Pig, , Data types in Pig and Running Pig, Oozie, Mahout,

Familiarization with standards: IEEE 2302-2021, IEEE P3652.1 IEEE 2410-2021, IEEE P7002, IEEE P7003, IEEE P2510.

Laboratory Work: Data Engineering Hadoop ecosystem, Spark etc.

Course Learning Outcomes (CLO):

The student will be able to:

1. Identify the issues and challenges related to Big Data
2. Design efficient algorithms for mining the data from large volumes.
3. Analyze the HADOOP and Map Reduce technologies.
4. Implement Big Data analytics techniques for various applications.

Text Books:

1. R. Shankarmani, M. Vijayalakshmi, “Big Data Analytic”, Wiley 2016
2. Anand Rajaraman and Jeffrey David Ullman, “Mining of Massive Datasets”, CUP,2012.

Reference Books:

1. Michael Minelli, Michelle Chambers, and AmbigaDhiraj, "Big Data, Big Analytics: Emerging
2. Business Intelligence and Analytic Trends for Today's Businesses", Wiley, 2013.